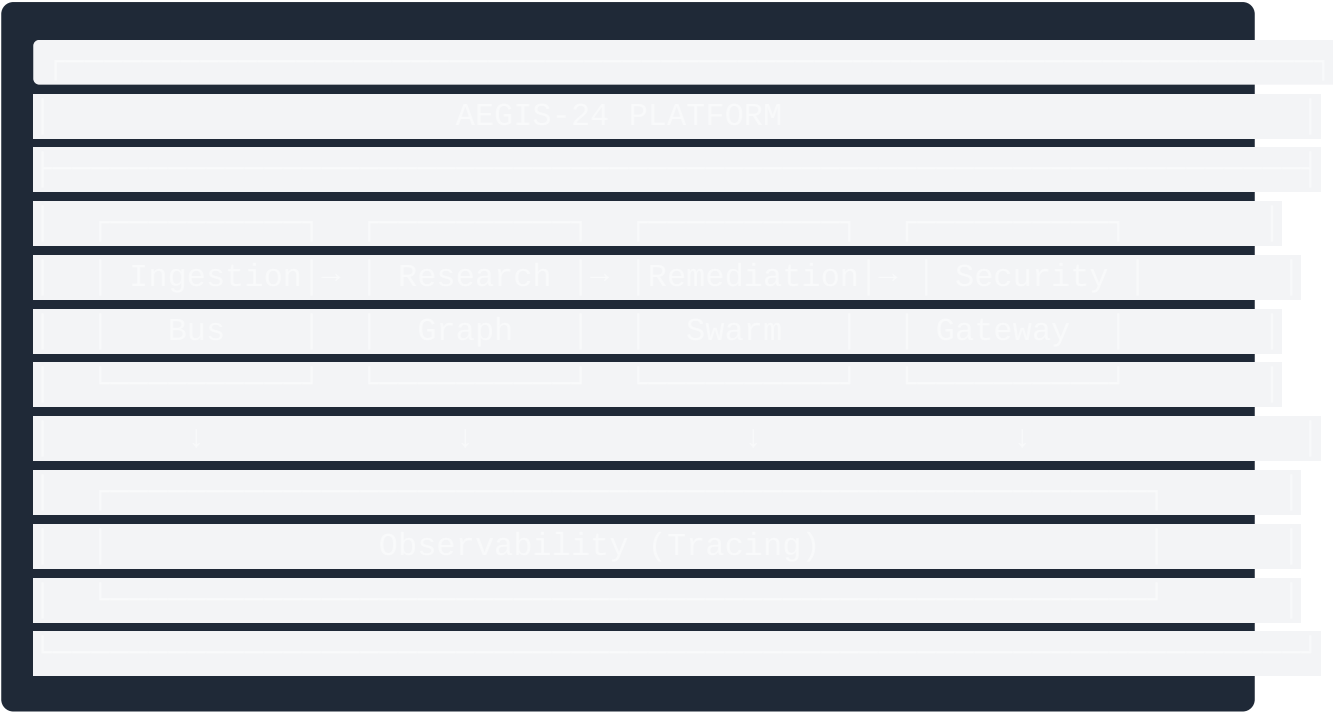


# Aegis-24 Workflow Guide

## System Architecture Overview

Aegis-24 is a production-hardened autonomous agent platform with five core components:



# Component Workflows

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## 1. Ingestion DAG (Data Flow)



### Key Features:

- Async event-driven processing
- Idempotency via Redis deduplication
- >1,000 events/sec throughput target
- MCP connector support for external data sources

## 2. Research State Machine (Cyclic Graph)



### Key Features:

- LangGraph-style stateful execution
- SQLite checkpointing at every node
- Reflexion loop capped at 3 iterations
- Quality score threshold: 0.8

### State Persistence:



### 3. Remediation Swarm (Hierarchical Process)



#### Key Features:

- CrewAI hierarchical agent swarm
- E2B micro-VM sandbox isolation
- Zero local code execution
- Automatic retry on test failure

4. Safety Gateway (Policy Enforcement Graph)



Key Features:

- Guardrails AI policy enforcement
- PII detection and redaction
- SEC/FINRA compliance checking
- Confidence-based escalation

5. Data Lineage Flow



### Tracking:

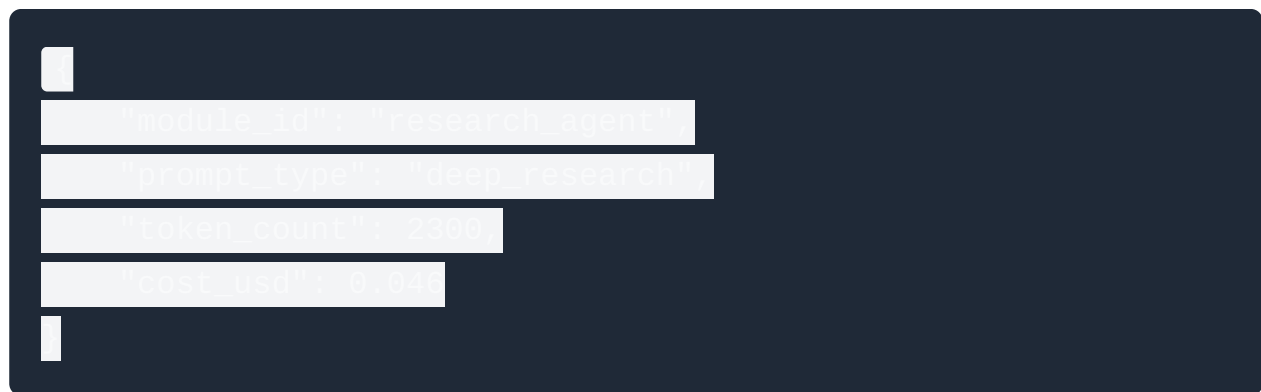
- Every transformation logged
- Source-to-destination traceability
- Timestamp recording
- Module attribution

## Token Management

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### Cost Attribution

Every LLM call emits trace metadata:



### Budget Enforcement

| Constraint                | Limit        |
|---------------------------|--------------|
| Reflexion Loop Iterations | Max 3        |
| Per-Request Token Budget  | Configurable |
| Per-Module Daily Budget   | Configurable |

## Context Isolation

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Financial data context and system error context are strictly separated:



## Eval Loop



### Security Gate (Binary):

- 0 PII leaks allowed
- 0 successful jailbreaks
- 0 sandbox escapes

## Orchestration Commands

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```
make setup-platform      # Initialize platform
make build-research-graph # Build research engine
make build-dev-swarm     # Build remediation swarm
make apply-guardrails    # Apply security policies
make instrument-traces   # Set up observability
make run-redteam         # Run adversarial testing
make generate-audit      # Generate compliance report
make push-to-github      # Deploy to GitHub
```

## Done State Criteria

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A task is only considered done when:

1. Eval Loop passes
2. Security gates cleared
3. Zero PII leaks
4. Zero jailbreaks
5. Sandbox escape prevented